

City Pollutant Reduction Plan: User Manual

City Pollutant Reduction Plan using Simplex method.
Program that computes optimal amount of
mitigation projects to implement while spending least
amount of money.

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Installation, and Pre-requisites Before Running

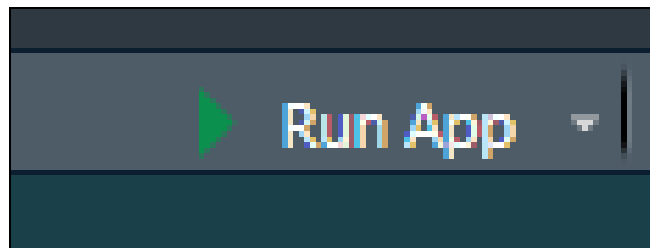
1. Before running the program, make sure you have the proper libraries installed (shiny, shinylibraries). Before starting the app, open **RStudio**, then in your console, type the following (*Take note, you will do this line by line and not like this where you will enter everything in one go*):

```
> install.packages("shiny")
install.packages("shinydashboard")
install.packages("bslib")
```

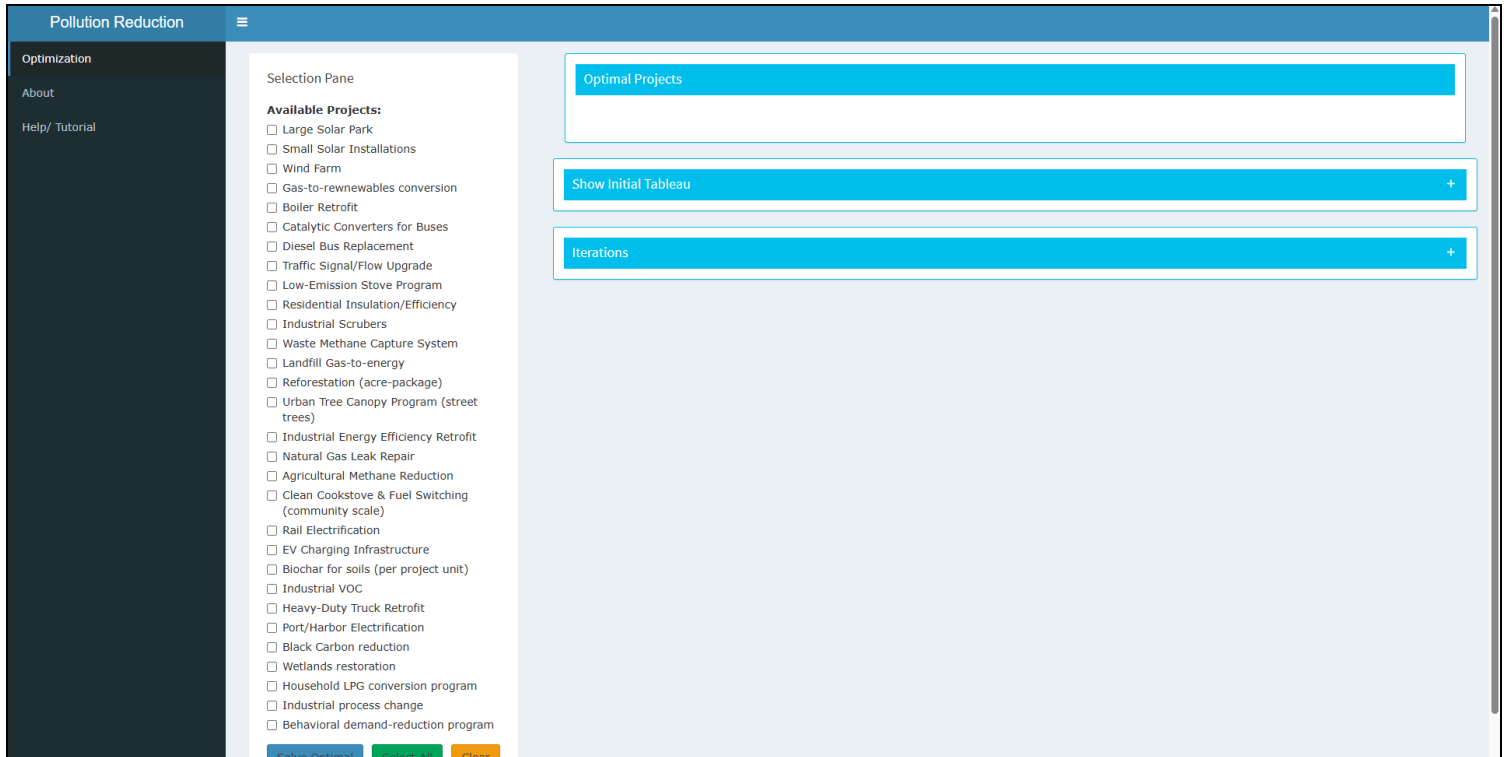
2. After installing the required packages, make sure that all of the extracted files from the .zip file are in one folder. There should be a total of 4 R files.

Name	Date modified	Type	Size
▼ Today			
 dataset	25/11/2025 8:29 AM	R File	3 KB
 simplex	25/11/2025 8:25 AM	R File	5 KB
 app	25/11/2025 8:11 AM	R File	8 KB
 optimal	25/11/2025 4:01 AM	R File	5 KB

3. Make sure that in R, you are working in the current directory where the files were extracted. Before starting the application, type **getwd()** in your console and confirm that you are in the correct working directory.
4. If all things are set, just click on the **Run App** button found on the upper right of your RStudio application. If errors occur, make sure that your R and RStudio are up to date, and you have installed the required libraries / packages.



Navigating Through the User Interface



Upon opening the application, this will be the first page you will see. Three blank cards containing **Optimal Projects**, **Show Initial Tableau**, and **Iterations**. On the left side are the checkboxes for the available mitigation projects. We have a panel on the left for tabs **Optimization**, **About**, and **Help / Tutorial**.

Panels on Optimization Tab

- **Optimal Projects** - Shows the needed mitigation projects, as well as the units required and the cost. Also shows whether your selected projects are feasible or not feasible, and if feasible, shows the total cost.
- **Show Initial Tableau** - Displays the initial tableau before going through any iterations.
- **Iterations** - Shows iterations. If not feasible, it shows up until the last iteration is computed.
- **Checkbox** - Contains all 30 mitigation projects, as well as buttons to solve for the optimal plan, select all mitigation options, and clear current selection.

1. In the selection pane, select all the projects that you want to implement.
2. Click the **Solve Optimal** button. For this example, let us say that **Select All** was clicked before solving the optimal solution.

Optimal Projects		
Your selection is feasible		
The cost of this optimal mitigation project is \$233966.99		
Project	Units	Cost
Boiler Retrofit	20.00	\$28000
Traffic Signal/Flow Upgrade	6.99	\$6993.23432100705
Low-Emission Stove Program	20.00	\$3600
Industrial Scrubbers	2.25	\$9431.45030368338
Reforestation (acre-package)	20.00	\$4400
Agricultural Methane Reduction	18.38	\$51455.0254266572
Clean Cookstove & Fuel Switching (community scale)	20.00	\$9000
Biochar for soils (per project unit)	20.00	\$28000
Industrial VOC	5.15	\$13383.1820221892
Wetlands restoration	20.00	\$36000
Household LPG conversion program	20.00	\$14000
Industrial process change	4.34	\$21704.0930567606
Behavioral demand-reduction program	20.00	\$8000

This will be the output of the **Optimal Projects** pane. It shows whether the selected inputs are feasible or not, as well as the total cost and corresponding mitigation projects to be used along with the units and cost of each.

Show Initial Tableau															
Initial Tableau															
S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12	S13	S14	S15	S16
60.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-1.00	0.00	0.00	0.00	0.00	0.00
18.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-1.00	0.00	0.00	0.00	0.00
55.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-1.00	0.00	0.00	0.00
25.00	1.00	0.20	0.10	1.50	0.50	2.00	0.05	0.01	0.30	0.00	0.00	0.00	-1.00	0.00	0.00
20.00	0.90	0.40	0.20	0.10	0.05	1.20	0.02	0.01	0.05	0.00	0.00	0.00	0.00	-1.00	0.00
30.00	2.80	0.60	0.80	0.00	0.50	5.00	0.01	0.05	0.02	0.00	0.00	0.00	0.00	0.00	-1.00
48.00	3.20	0.90	1.00	0.00	0.70	6.00	0.02	0.08	0.03	0.00	0.00	0.00	0.00	0.00	0.00
12.00	0.60	0.10	0.40	0.05	0.20	3.00	0.02	0.02	0.01	0.00	0.00	0.00	0.00	0.00	0.00

If you click the **+** icon on **Show Initial Tableau** and **Iterations**, it will show the initial tableau of the selected projects, as well as the tableau per iteration.

If you wish to compute for another set of projects, just select your choices in the Selection Pane, and then click on the **Solve Optimal** button. Upon clicking, the panels will update with the new set of answers.

Optimal Projects

Your selection is **infeasible**

Number of iterations before ending: **31**

Selection Pane

Available Projects:

- ☒ Large Solar Park
- ☒ Small Solar Installations
- ☒ Wind Farm
- ☒ Gas-to-renewables conversion
- ☒ Boiler Retrofit
- ☒ Catalytic Converters for Buses
- ☒ Diesel Bus Replacement
- ☒ Traffic Signal/Flow Upgrade
- ☒ Low-Emission Stove Program
- ☒ Residential Insulation/Efficiency
- ☒ Industrial Scrubbers
- ☒ Waste Methane Capture System
- ☒ Landfill Gas-to-energy
- ☒ Reforestation (acre-package)
- ☒ Urban Tree Canopy Program (street trees)
- ☒ Industrial Energy Efficiency Retrofit
- ☒ Natural Gas Leak Repair
- ☒ Agricultural Methane Reduction
- ☒ Clean Cookstove & Fuel Switching (community scale)
- ☒ Rail Electrification
- ☒ EV Charging Infrastructure
- ☒ Biochar for soils (per project unit)
- ☒ Industrial VOC
- ☒ Heavy-Duty Truck Retrofit
- ☐ Port/Harbor Electrification
- ☐ Black Carbon reduction
- ☐ Wetlands restoration
- ☐ Household LPG conversion program
- ☐ Industrial process change
- ☐ Behavioral demand-reduction program

Show Initial Tableau

Initial Tableau

S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12	S13
60.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-1.00	0.00	0.00
18.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-1.00	0.00
55.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-1.00
25.00	1.00	0.20	0.10	1.50	0.50	2.00	0.05	0.01	0.30	0.00	0.00	0.00
20.00	0.90	0.40	0.20	0.10	0.05	1.20	0.02	0.01	0.05	0.00	0.00	0.00
30.00	2.80	0.60	0.80	0.00	0.50	5.00	0.01	0.05	0.02	0.00	0.00	0.00
48.00	3.20	0.90	1.00	0.00	0.70	6.00	0.02	0.08	0.03	0.00	0.00	0.00
12.00	0.60	0.10	0.40	0.05	0.20	3.00	0.02	0.02	0.01	0.00	0.00	0.00
2.00	0.02	0.01	0.70	0.00	0.01	1.50	0.03	0.20	0.00	0.00	0.00	0.00
15.00	0.10	0.05	0.05	0.02	0.02	0.50	0.00	0.00	0.01	0.00	0.00	0.00
6.00	0.40	6.00	0.40	0.00	0.10	0.60	0.01	0.01	0.00	0.00	0.00	0.00
28.00	0.20	0.10	0.05	8.00	0.20	0.10	0.00	0.00	0.05	0.00	0.00	0.00
24.00	0.15	0.05	0.03	6.50	0.10	0.05	0.00	0.00	0.03	0.00	0.00	0.00
3.50	0.04	0.02	0.01	0.80	0.03	0.10	0.01	0.00	0.00	0.00	0.00	0.00
4.20	0.06	0.01	0.03	0.60	0.02	0.15	0.00	0.02	0.00	0.00	0.00	0.00
22.00	0.50	0.30	0.15	0.20	0.10	1.00	0.01	0.01	0.03	0.00	0.00	0.00
10.00	0.05	0.01	0.01	4.00	0.02	0.02	0.00	0.00	0.01	0.00	0.00	0.00
8.00	0.02	0.01	0.02	7.20	0.05	0.02	0.10	0.00	0.05	0.00	0.00	0.00
3.20	0.04	0.02	0.90	0.10	0.02	2.00	0.05	0.25	0.00	0.00	0.00	0.00
80.00	2.00	0.40	1.20	0.00	0.60	10.00	0.02	0.10	0.05	0.00	0.00	0.00
20.00	0.30	0.05	0.10	0.00	0.05	0.50	0.01	0.01	0.01	0.00	0.00	0.00